AMANLEB

AI-Powered Scam & Smishing Detection for Lebanon

LebNet AI Internship - Final Project Report

Prepared by	Mohammad Naim
Project	AmanLeb
Program	LebNet AI Internship
Date	20 August 2026

A multilingual AI safety prototype combining classification, retrieval, and grounded cybersecurity guidance.

Abstract

AmanLeb is a multilingual SMS safety prototype designed to detect ham, spam, and smishing messages and provide trusted, source-grounded guidance for suspicious cases. The project was developed as a LebNet AI Internship final project with a focus on practical machine learning, multilingual Natural Language Processing, Transformer fine-tuning, retrieval-augmented systems, and safe user-facing AI design. The project began with cleaning and exploratory analysis of a 5,971-row SMS phishing dataset. After normalizing labels and removing conflicting and duplicate messages, the final dataset contained 5,947 unique SMS texts. A TF-IDF plus class-weighted Logistic Regression baseline achieved 96.08% test accuracy and a macro F1 score of 0.884. A weighted multilingual DistilBERT classifier improved performance to 97.3% accuracy and 0.905 macro F1, while reducing suspicious-to-ham errors from five to two on the held-out test set. AmanLeb then adds a confidence-based safety-routing layer. High-confidence ham predictions bypass retrieval, while low-confidence ham, spam, and smishing cases trigger multilingual semantic retrieval from trusted Lebanese cybersecurity sources. Sentence Transformers and ChromaDB are used for retrieval, and the final runtime employs deterministic extractive guidance instead of unconstrained generation. A Qwen-based generation experiment was evaluated but was not retained in the deployed runtime because it could introduce unsupported claims or violate grounding constraints. Six end-to-end challenge tests across English, Arabic, French, and English-Arabizi inputs confirmed stable behavior after code cleanup. The resulting prototype demonstrates a safety-oriented architecture that separates classification, uncertainty routing, evidence retrieval, and user guidance while clearly documenting current limitations and future work.

Key result

The final weighted multilingual DistilBERT achieved 97.3% test accuracy, 0.905 macro F1, and only 2 suspicious-to-ham errors on the 893-message held-out test split.

Contents

1. Introduction

1.1 Background

SMS scams remain a practical cybersecurity problem because attackers can imitate familiar organizations, create urgency, and request clicks, credentials, personal information, or payments in only a few lines of text. The problem becomes more challenging in multilingual environments such as Lebanon, where a user may receive messages in English, Arabic, French, or mixed forms such as English combined with Arabizi.

AmanLeb addresses this problem as an educational AI prototype. Instead of relying on a single prediction, the system combines a supervised text classifier with an additional safety-routing layer and trusted-source retrieval. This architecture is designed to distinguish between what the model predicts and what safety guidance is shown to the user.

1.2 Problem Statement

A practical SMS safety system must do more than achieve high aggregate accuracy. It must minimize dangerous false negatives, remain useful across languages, avoid presenting unsupported advice, and communicate uncertainty when the classifier is not sufficiently confident. In this project, suspicious messages incorrectly classified as ham are therefore treated as a particularly important error type.

1.3 Project Objectives

- Clean and analyze an SMS phishing dataset containing ham, spam, and smishing examples.
- Establish a strong, interpretable classical machine-learning baseline.
- Fine-tune a multilingual Transformer and compare it fairly with the baseline using the same data split.
- Add a safety-routing layer that treats uncertain ham predictions conservatively.
- Retrieve relevant cybersecurity guidance from trusted Lebanese sources.
- Prefer grounded extractive guidance over unsupported generated advice.
- Integrate the full pipeline into an accessible Streamlit application.
- Validate the final implementation using multilingual and mixed-language challenge cases.

1.4 Main Contributions

The final project contributes a complete experimental and application pipeline rather than a single model. Its main components are summarized below.

Component	Purpose
EDA and cleaning	Build a reliable modeling dataset and document class imbalance and metadata quality.
Classical baseline	Provide an interpretable TF-IDF + Logistic Regression reference.
Multilingual DistilBERT	Improve contextual and multilingual SMS classification.
Safety routing	Escalate uncertain ham and all suspicious predictions to additional analysis.
RAG	Retrieve evidence from trusted Lebanese cybersecurity sources.
Extractive guidance	Provide traceable advice without fabricating unsupported claims.
Streamlit app	Expose the full pipeline through a simple interactive interface.

2. Dataset and Exploratory Data Analysis

2.1 Raw Dataset

The project uses the Mendeley dataset "SMS PHISHING DATASET FOR MACHINE LEARNING AND PATTERN RECOGNITION". The raw CSV contained 5,971 rows and five columns: LABEL, TEXT, URL, EMAIL, and PHONE.

Field	Description
LABEL	Message class: ham, spam, or smishing, with inconsistent capitalization in the raw data.
TEXT	SMS message content.
URL	Provided binary indicator for URL presence.
EMAIL	Provided binary indicator for email presence.
PHONE	Provided binary indicator for phone-number presence.

2.2 Label Normalization and Cleaning

The raw class labels appeared as ham, Smishing, smishing, spam, and Spam. Labels were converted to lowercase and stripped of surrounding whitespace. Binary metadata fields were also standardized.

Duplicate analysis identified 17 duplicated full rows and 22 duplicated TEXT entries beyond the first occurrence. Two unique texts had conflicting spam/smishing labels, corresponding to four problematic rows. Those conflicting examples were removed, followed by removal of remaining duplicated message texts.

Clean dataset
Final size: 5,947 rows and 5,947 unique SMS texts. No missing or invalid target labels remained.

2.3 Class Distribution

Class	Approx. Share
Ham	81.28%
Smishing	10.53%
Spam	8.19%

The strong class imbalance means that accuracy alone can hide weak minority-class behavior. This directly motivated the use of macro F1 and safety-oriented confusion-matrix analysis.

2.4 Message Length and Metadata Findings

EDA indicated that ham messages are generally shorter, while suspicious messages tend to be longer on average. However, the distributions overlap substantially, so message length alone is not a reliable discriminator and no aggressive length-based outlier removal was applied.

The supplied metadata indicators were informative but noisy. Their conditional frequencies by class were:

Feature	Ham	Smishing	Spam
URL = yes	0.2%	15.7%	20.1%
EMAIL = yes	0.0%	1.8%	1.4%
PHONE = yes	0.1%	82.4%	63.7%

Because these fields can contain extraction noise and are not always available in real-world input, the classical baseline was intentionally built as a text-only model.

3. Experimental Design and Evaluation Strategy

3.1 Shared Data Split

To compare models fairly, the same stratified split was used for both the classical and Transformer experiments. The split used random_state=42.

Split	Share	Messages
Train	70%	4,162
Validation	15%	892
Test	15%	893

The validation set was used for model selection and tuning decisions. The test set was kept untouched until the final evaluation of each selected model.

3.2 Metrics

- Accuracy: overall fraction of correct predictions.
- Precision: reliability of positive predictions for each class.
- Recall: fraction of each true class successfully recovered.
- F1-score: harmonic balance between precision and recall.
- Macro F1: unweighted average of the per-class F1 scores, giving minority classes equal importance.
- Confusion matrix: direct inspection of which classes are confused with one another.

3.3 Safety-Oriented Error Analysis

For this application, suspicious messages predicted as ham are more concerning than confusing spam with smishing. Two additional counts were therefore monitored:

- Suspicious -> Ham: all true spam or smishing messages predicted as ham.
- Smishing -> Ham: true smishing messages predicted specifically as ham.

Interpretation rule
A low suspicious-to-ham count is desirable, but a zero count on one held-out test split is not a guarantee of zero misses in deployment.

4. Classical Baseline: TF-IDF + Logistic Regression

4.1 TF-IDF Representation

The baseline represents SMS messages with TfidfVectorizer. The vectorizer is fit only on the training text and then used to transform validation and test sets, preventing information leakage. The resulting vocabulary contained 7,843 TF-IDF features.

4.2 Baseline and Class-Weighted Models

An unweighted Logistic Regression model first achieved 95.07% validation accuracy and approximately 0.863 macro F1, but it produced 31 suspicious-to-ham errors. Because of the severe class imbalance, a second Logistic Regression model was trained with class_weight="balanced".

The balanced model improved the validation result to 96.30% accuracy, approximately 0.891 macro F1, and only 10 suspicious-to-ham errors. It was therefore selected for final testing.

4.3 Final Logistic Regression Test Results

Metric	Result
Accuracy	96.08%
Macro F1	0.884
Suspicious -> Ham	5
Smishing -> Ham	0

Actual / Predicted	Ham	Spam	Smishing
Ham	717	4	5
Spam	5	61	7
Smishing	0	14	80

The balanced classical model became a strong reference baseline. Its coefficient-based interpretability also provided insight into which TF-IDF terms were associated with each class.

5. Multilingual Transformer Classifier

5.1 Model Choice

The Transformer stage uses distilbert/distilbert-base-multilingual-cased, fine-tuned for three-class sequence classification. The label mapping is fixed as ham=0, spam=1, and smishing=2. The full model is fine-tuned rather than using frozen embeddings.

5.2 Tokenization and Sequence Length

The multilingual WordPiece tokenizer supports multiple writing systems. A token-length audit showed that only six training messages exceeded 128 tokens. MAX_LENGTH=128 was therefore selected to retain nearly all SMS content while reducing memory and computation. Dynamic padding was used during training.

5.3 Unweighted Transformer

The first fine-tuning experiment used standard cross-entropy loss. Its best validation checkpoint occurred at epoch 2, achieving approximately 96.19% validation accuracy, 0.877 macro F1, and 8 suspicious-to-ham errors.

5.4 Weighted Transformer

A second model was trained from a fresh pretrained checkpoint using class weights computed from the training split. Weighted cross-entropy increased the contribution of minority-class errors during optimization. The best validation checkpoint occurred at epoch 3.

Validation Metric	Weighted Transformer
Accuracy	96.86%
Macro F1	0.902
Suspicious -> Ham	5
Smishing -> Ham	0

5.5 Final Transformer Test Results

Class	Precision	Recall	F1	Support
Ham	0.997	1.000	0.999	726
Spam	0.836	0.836	0.836	73
Smishing	0.891	0.872	0.882	94

Overall Metric	Result
Accuracy	97.3%
Macro Precision	0.908
Macro Recall	0.903
Macro F1	0.905
Suspicious -> Ham	2
Smishing -> Ham	0

Actual / Predicted	Ham	Spam	Smishing
Ham	726	0	0
Spam	2	61	10
Smishing	0	12	82

Model selection
The weighted multilingual DistilBERT outperformed the balanced Logistic Regression in both macro F1 and suspicious-to-ham errors, so it was selected as AmanLeb's final classifier.

6. Retrieval-Augmented Safety Guidance

6.1 Purpose

The retrieval stage is not a second classifier. Its purpose is to provide relevant, traceable cybersecurity evidence and user guidance after the classifier and safety-routing layer decide that additional review is appropriate.

6.2 Trusted Sources

Source	Organization	Role
ISF fraudulent SMS warning	Lebanese Internal Security Forces	Specific warning about fraudulent SMS impersonating a money-transfer company.
ISF Internet Security Awareness	Lebanese Internal Security Forces	Broad internet and account-security guidance.
Alfa Security Tips	Alfa	General telecom and account-security recommendations.

6.3 Retrieval Pipeline

Each trusted webpage is downloaded and converted to visible text. Source-specific start/end markers remove navigation and unrelated page sections. The cleaned text is split into overlapping 80-word chunks with 20-word overlap.

Each chunk is embedded with sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2, producing a 384-dimensional multilingual vector. ChromaDB stores the chunk text, vector, and provenance metadata such as title, organization, URL, source ID, and chunk index.

SMS requiring additional safety analysis
Multilingual query embedding
Chroma semantic retrieval
Top trusted-source chunk
Append adjacent same-source chunk when available
Strict extractive explanation and actions
Official source attribution and link

6.4 Qwen Grounded-Generation Experiment

Qwen/Qwen2.5-1.5B-Instruct was evaluated as an optional generator that could turn retrieved evidence into a short explanation. The experiment revealed an important engineering limitation: fluent output was not consistently grounded. The model could add unsupported risks, ignore required formatting, or fail citation constraints.

A deterministic validator was developed to reject unsupported outputs. The final decision was to avoid requiring Qwen in the Streamlit runtime. The deployed prototype instead uses deterministic extractive analysis from retrieved official evidence.

Safety principle
When the retrieved evidence does not contain a sufficiently specific explanation or action, AmanLeb says so instead of inventing one.

6.5 Safe Extractive Analysis

The final build_safe_analysis pipeline retrieves the top chunk, appends the next chunk from the same source when available, normalizes whitespace, searches for a sufficiently strong fraud-related sentence, and extracts up to three action statements beginning with trusted-source phrases such as "Do not click", "Do not enter", and "Do not forward".

7. Streamlit Application and Safety Routing

7.1 User Interface

The final Streamlit interface accepts a user SMS, displays the Transformer prediction, shows ham/spam/smishing probabilities, assigns a user-facing safety status, and conditionally runs trusted-source retrieval. The classifier is cached at startup, while the RAG engine is loaded lazily only when needed.

7.2 Safety Status Logic

Condition	Safety Status	RAG?
Predicted ham and P(ham) >= 0.80	Likely Safe	No
Predicted ham and P(ham) < 0.80	Needs Review	Yes
Predicted spam or smishing	Suspicious	Yes

The 0.80 threshold is a conservative prototype heuristic, not a statistically optimized or calibrated production threshold. It was introduced to reduce the risk of immediately presenting a moderate-confidence ham prediction as safe.

7.3 Separation of Responsibilities

Layer	Responsibility
Transformer classifier	Predict ham, spam, or smishing and produce class probabilities.
Safety router	Convert model output and ham confidence into Likely Safe, Needs Review, or Suspicious.
RAG retriever	Retrieve relevant trusted-source evidence.
Extractive guidance	Present only source-grounded explanation/actions when specific evidence is available.
Streamlit UI	Expose results, confidence, safety status, actions, and official source links.

8. Results and End-to-End Validation

8.1 Final Model Comparison

Model	Test Accuracy	Macro F1	Suspicious -> Ham	Smishing -> Ham
Balanced Logistic Regression	96.08%	0.884	5	0
Weighted Multilingual DistilBERT	97.3%	0.905	2	0

The Transformer improved both aggregate performance and the safety-oriented suspicious-to-ham error count. The largest remaining classification difficulty is the boundary between spam and smishing, which is less safety-critical than suspicious-to-ham confusion but still important for taxonomy quality.

8.2 Six Final Challenge Tests

Test	Observed Result	Assessment
Normal English HAM	HAM 99.6% - Likely Safe	Strong pass; RAG correctly skipped.
Promotional prize message	SMISHING 88.0% - Suspicious	Suspicious detection passed; exact spam/smishing class confused.

Test	Observed Result	Assessment
English banking scam	SMISHING 98.7% - Suspicious	Strong classification pass; retrieval worked, extraction limited.
Arabic banking scam	SMISHING 98.0% - Suspicious	Strong multilingual classification and cross-lingual retrieval.
French banking scam	SMISHING 98.4% - Suspicious	Strong multilingual classification and cross-lingual retrieval.
English + Arabizi scam	SMISHING 98.7% - Suspicious	Strong mixed-language classification; retrieval worked.

All six tests were repeated after source-code cleanup and reproduced the same outcomes, confirming that the refactoring changed code organization rather than application behavior.

8.3 Observed RAG Limitation

The final tests revealed a systematic but safe limitation: semantic retrieval often returns a broadly relevant trusted source, while the strict extractor may fail to find a sufficiently specific fraud sentence or action in the selected chunk. In these cases, AmanLeb deliberately shows a "no sufficiently specific explanation" message rather than generating unsupported guidance.

9. Implementation and Reproducibility

9.1 Final Project Structure

The repository is organized into application code, reusable source code, data, model artifacts, notebooks, screenshots, and report files. The four development notebooks are numbered in experimental order.

9.2 Main Runtime Components

File / Directory	Purpose
app/app.py	Streamlit interface, Transformer inference, confidence routing, conditional RAG display.
src/rag.py	Trusted-source extraction, chunking, Chroma persistence, retrieval, and safe extractive analysis.
models/amanleb_final_transformer/	Saved Transformer config, tokenizer, and model weights.
models/chroma_db/	Persistent Chroma vector database.
notebooks/	EDA, Logistic Regression, Transformer, and RAG development notebooks.
requirements.txt	Python dependencies needed to reproduce the project environment.
.gitignore	Excludes virtual environments, caches, and runtime-generated files from Git.

9.3 Reproducibility Notes

- The classical and Transformer experiments reuse the same stratified 70/15/15 split with random_state=42.
- The final Transformer label mapping is stored in config.json.
- The application uses MAX_SEQUENCE_LENGTH=128, matching the fine-tuning setup.
- The RAG embedding model is fixed to paraphrase-multilingual-MiniLM-L12-v2.
- The Chroma database is persistent and is reused when already populated.
- The final source code was re-tested on the six challenge messages after cleanup.

9.4 Running the Application

From the project root, install dependencies from requirements.txt, ensure the fine-tuned model folder is present, and launch:

```
python -m streamlit run app/app.py
```

10. Limitations and Safety Considerations

- The labeled training dataset is relatively small and is not specifically collected from Lebanon.
- Spam and smishing can overlap semantically, and the model still confuses these classes in some cases.
- The six challenge tests are qualitative validation cases, not a replacement for a large external benchmark.
- The current RAG knowledge base contains only three trusted Lebanese sources.
- Top-1 retrieval can choose a broad source instead of the most specific fraud warning.
- Strict extraction can return no explanation/actions even when retrieval is broadly relevant.
- The 0.80 ham-confidence threshold is heuristic and has not been formally calibrated.
- New scam campaigns, dialects, Arabizi spellings, shortened URLs, or adversarial messages may produce different behavior.
- AmanLeb is an educational prototype and should not be treated as a guarantee that a message is safe or malicious.

10.1 Responsible Use

For banking credentials, payment information, personal data, account access, or urgent financial requests, users should independently verify the message through the organization's official communication channels. The project is best understood as decision-support assistance rather than an automated authority.

11. Future Work

- Collect a larger Lebanon-specific multilingual SMS dataset, including Arabic dialect and Arabizi.
- Evaluate additional classical models such as Decision Trees and other baselines.
- Expand the trusted-source knowledge base and introduce automatic source refresh mechanisms.
- Use top-k retrieval with reranking to improve evidence specificity.
- Improve extractive evidence selection while maintaining strict provenance.
- Calibrate the safety-routing threshold on a dedicated validation objective.
- Evaluate stronger grounded generation with more robust citation and hallucination controls.
- Add a conversational GPT/API interface as a separate layer while preserving classifier/RAG separation.
- Explore dedicated agentic tool use for verified source lookup and reporting.
- Perform larger external, multilingual, adversarial, and robustness evaluations.

12. Conclusion

AmanLeb demonstrates an end-to-end AI workflow for multilingual SMS safety. The project began with careful data cleaning and EDA, established a strong class-weighted Logistic Regression baseline, and then improved performance with a weighted multilingual DistilBERT classifier. On the held-out test set, the final Transformer achieved 97.3% accuracy and 0.905 macro F1 while reducing suspicious-to-ham errors to two.

The project also goes beyond classification. A confidence-based routing layer prevents moderate-confidence ham predictions from automatically being treated as safe, while multilingual semantic retrieval provides traceable evidence from official Lebanese sources. Experiments with Qwen highlighted the importance of grounding and validation; the final runtime therefore prioritizes deterministic extractive guidance over unconstrained generation.

The final Streamlit application integrates these components into a coherent prototype that was revalidated on six English, Arabic, French, and mixed-language challenge messages after source-code cleanup. Although the system has important limitations, especially in evidence coverage and spam/smishing boundary quality, the project provides a clear foundation for future Lebanon-specific scam detection, safer RAG, and richer conversational assistance.

Final takeaway
AmanLeb is not only a classifier. It is a layered safety prototype that separates prediction, uncertainty handling, trusted evidence retrieval, and user guidance.

Appendix A. Final Project Structure

```
amanleb/
|-- app/
|   `-- app.py
|-- data/
|   |-- raw/Dataset_5971.csv
|   `-- processed/Dataset_5971_clean.csv
|-- models/
|   |-- amanleb_final_transformer/
|   |   |-- config.json
|   |   |-- model.safetensors
|   |   |-- tokenizer.json
|   |   `-- tokenizer_config.json
|   `-- chroma_db/
|-- notebooks/
|   |-- 01_data_exploration.ipynb
|   |-- 02_logistic_regression.ipynb
|   |-- 03_transformer_classifier.ipynb
|   `-- 04_rag.ipynb
|-- results/screenshots/
|-- report/
|-- src/
|   |-- __init__.py
|   `-- rag.py
|-- .gitignore
|-- requirements.txt
`-- README.md
```

Appendix B. Final Challenge Test Messages

1. Normal English HAM

Hey, are we still meeting at university tomorrow at 10? I will be there a few minutes early.

2. Promotional prize message

Congratulations! You have won a FREE shopping voucher worth $500. Reply WIN now to claim your prize. Limited offer!

3. English banking smishing

BLOM Bank Security Alert: Your account has been temporarily blocked. Verify your banking details immediately at http://secure-blom-account.com to restore access.

4. Arabic banking scam

تم تعليق حسابك المصرفي مؤقتاً. يرجى الضغط على الرابط التالي وتحديث معلومات حسابك لتجنب إغلاق الحساب: http://secure-bank-update.com

5. French banking scam

Alerte de sécurité bancaire : Votre compte a été temporairement suspendu. Cliquez sur le lien suivant et confirmez vos informations bancaires pour réactiver votre compte : http://secure-bank-verification.com

6. English + Arabizi scam

Hi, your OMT payment is still pending. Please confirm your details today la ma yensakar el account: http://omt-payment-check.com

References

1. Mendeley Data. "SMS PHISHING DATASET FOR MACHINE LEARNING AND PATTERN RECOGNITION." Dataset used for the AmanLeb experiments.

2. Hugging Face. distilbert/distilbert-base-multilingual-cased. Pretrained multilingual DistilBERT model.

3. Sentence Transformers. sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2. Multilingual embedding model.

4. ChromaDB. Persistent vector database used for semantic retrieval.

5. Lebanese Internal Security Forces. Fraudulent text messages impersonating a money-transfer company - warning on theft of personal data and money. https://isf.gov.lb/news/fraudulent-text-messages-impersonating-a-money-transfer-company-beware-of-the-theft-of-your-personal-data-and-money/

6. Lebanese Internal Security Forces. Internet Security Awareness. https://isf.gov.lb/internet-security-awareness/

7. Alfa. Security Tips. https://www.alfa.com.lb/en/support/security-tips

8. Qwen. Qwen/Qwen2.5-1.5B-Instruct. Evaluated experimentally for grounded response generation; not required by the final runtime.